

BS Computer Science

Examination: Final 2023

Semester: 6th

Course Code: CS3201

Title: Computer Network

Marks: 40

Time allowed: 155 minutes

Subjective

Attempt any Four Questions

Question 2 10 marks]

- a. Make a clear comparison between:
i. TCP and UDP 5 ii. Persistent http and non-persistent http 5

Question 3 10 marks]

- a. Explain Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM). 5
b. State two defending points for circuit switching in contrast to packet switching. 5

Question 4 10 marks]

- a. Elaborate each layer of OSI Model. 7
b. Explain LAN and WAN briefly. 3

Question 5 10 marks]

- a. What is Automatic Repeat Request (ARQ)? 3
b. Elaborate working principle of ARQ. 3
c. State the difference between Stop-and-Wait ARQ and Go-Back-N ARQ. 4

Question 6 10 marks]

- a. What is transmission mode? Discuss its types. 5
b. Discuss two error detection techniques. 5

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Q.No.1:

Consider the following bit sequence: **01101001**.

Sketch coding techniques transmitted signal amplitude against time for the following cases:

- a) NRZ
- b) NRZ-I
- c) Manchester Coding

Q.No.2:

Given IP Address: **192.168.10.0/28**

- a) Find the subnet mask?
- b) Find the total number of networks that can be created?
- c) Find the total number of IP Addresses on each network?
- d) Find the total number of Hosts on each network?
- e) Find Network ID, Host IP Address & Broadcast ID?

Q.No.3:

A bit stream **10011101** is transmitted using the standard CRC method. The generator polynomial is x^3+1 .

- a) What is the actual bit string transmitted?
- b) Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?

Q.No.4:

The Physical Layer cannot accommodate an arbitrary number of users. Describe the techniques that will help to resolve this problem.

Q.No.5:

What is meant by Protocol? Describe the protocols in detail that are used in the **3rd Layer of OSI Model**.
